

Questions and answers for the judges

What each member will be asked and what to answer, then a full set for whoever presents.
Written in plain words: no answer here needs a technical background to say or to understand.

READ THIS FIRST

The whole project in six sentences

Anyone on the team should be able to say this without notes.

1. Two things are needed in forensics and they pull in opposite directions: sometimes you must destroy data so nobody can ever recover it, and sometimes you must recover data that somebody tried to destroy.
 2. Today those are sold as separate products, so the two halves of a case never share one record.
 3. We built one tool that does both, and writes every action into a single register where each entry is locked to the one before it.
 4. Remove a page from that register and the tool does not just say “invalid”. It says *which page* is missing.
 5. A second computer, held by a second person, signs every action, so one operator alone cannot produce a record that looks complete.
 6. It prints the two-signature certificate Indian courts have required since July 2024, and it refuses to claim more than it actually did.

The numbers, all of them checked

FIGURE	WHAT IT MEANS
630 tests pass, none skipped	every claim we make is checked by an automatic test
11 of 11 automated checks green	the full check suite, run end to end
8 of 8 claims survived	we tried to disprove our own claims and could not
4 of 4 files returned byte for byte	recovery is exact, or we do not claim it
3 separate checkers agree	written in three languages so one shared mistake cannot hide
37 thousandths of a second	time to find a tampered record among 500
0 serious security findings	from an automated security scan of our own code

THE THREE THINGS WE MUST NEVER CLAIM

We do not claim the highest trust level. We reach the second highest. The highest needs a presence device we designed but did not build.

We do not claim the strongest wipe grade on pen drives. Flash memory keeps blocks no computer can address. We report the honest lower grade.

We do not claim the record proves nothing was hidden. It proves nothing was *changed*. No register in the world can prove what was never written down.

What each member will be asked

Judges often split up and question people separately. Each person should know their own section cold, and the six sentences above.

PULKIT · TEAM LEAD AND CORE ENGINEERING

Owns the whole system, the cryptography, and the final answer to anything nobody else can field.

Q What exactly did you build?

One tool that securely erases data and recovers deleted files, and writes every action into a single register where each entry carries the fingerprint of the one before it. A second computer co-signs each action.

Q Why is that hard? Isn't it just hashing?

Hashing a file is easy. Three things are not. **One:** making the tool refuse to run when the second party is missing, instead of quietly recording something weaker. **Two:** doing that check *before* the destructive step, because wiping cannot be undone. **Three:** making sure the record can never say more than actually happened.

Q How much of this is your own work?

The forensic tool is new. It builds on AetherProof, my own open-source project published before this problem statement was released. We declare that openly on the last slide rather than have it discovered.

Q What does not work yet?

The strongest wipe grade on solid-state drives. The code is written and tested, but proving it needs a drive we are allowed to destroy, and we could not obtain one. We marked it **not tested** rather than claim it.

"We would rather show you a gap than have you find one."

Owns the screens: the register view, the certificate, and the live integrity check.

Q Is the screen just showing pre-written results?

No. The page recalculates every fingerprint itself, in the browser, using the browser's own cryptography. It never asks a server whether the record is valid.

"Press F12 and you can watch each fingerprint being computed on your own machine."

Q Why do the check in the browser at all?

Because if our program says "this is valid", you are trusting us. When the page recomputes it in front of you, you are trusting arithmetic.

Q What happens when you press Inject Tamper?

One signed field is changed in memory only. Every fingerprint is then recomputed. That record no longer matches, and every record after it fails too, because each one is locked to the one before. The file on disk is never touched.

Q What did you build it with?

React and TypeScript, built with Vite, styled with Tailwind. It has no network calls at all and no security warnings in its dependencies.

Owns the 630 tests, the automated check suite, and the evidence that the numbers are real.

Q How do we know the tests actually test something?

Two ways. First, none are skipped: a skipped test is never counted as a pass anywhere in this project. Second, we ran an adversarial audit that actively tries to disprove each of our claims. Eight claims, eight survived.

Q Have you ever found a real bug?

Yes, and it is the best thing we can tell you. Hours before the deadline we found the recovery tool producing a broken archive while reporting it as complete. 620 tests passed anyway, because they only covered the three file types where the problem could not occur. We fixed it, added ten tests, and it now returns four of four files perfectly.

"A team that finds its own bugs is worth more than one that never reports any."

Q Does it work on a fresh machine, or only yours?

We extracted a clean copy into an empty folder and ran everything against it. That caught a real fault: on Windows the line endings changed on checkout and four tests failed. Fixed, and re-verified.

Owns the deck, the demo flow, and the room. Full question set in Part three.

Q Show me it working, in two minutes.

Register view, point at the refused row. Then the live check, press Inject Tamper, it names the exact record. Then Restore. Then the certificate with its stated limits. That is the whole demo and it needs no internet.

Q What is the single most important thing on screen?

The refused row. It is an operation that *did not happen*, recorded permanently.

“Ask any other tool to show you the log entry for a job it refused to perform. There isn’t one.”

Owns the mapping to Indian law and to the international standard.

Q Which law makes this necessary?

Section 63(4) of the Bharatiya Sakshya Adhiniyam 2023, in force since July 2024. It requires a certificate for electronic evidence signed by both the person in charge of the device and an expert. Our certificate has both parts, and each field is marked with the clause it answers.

Q And the sanitization side?

NIST Special Publication 800-88 Revision 2, finalised 26 September 2025. It defines the wipe grades and, new in Revision 2, a formal validation step. We implement the grades honestly and never record a stronger one than achieved. There is also the Digital Personal Data Protection Act 2023, section 8(7), which requires organisations to erase personal data.

Q Is this admissible in court?

That is for the court, not for us. What we can say is that the certificate contains what the section requires, field by field, with each clause named.

“We claim the form is right. We do not claim the verdict.”

Q You mention outside timestamping. Is that legally recognised?

Not in Indian courts, and we do not say it is. It provides the same class of cryptographic guarantee that courts increasingly accept. Nothing stronger than that.

Owns the competitor picture, the market numbers, and the laboratory relationship.

Q Who else does this, and why are you different?

Blancco and BitRaser are government-certified wiping tools; each certificate is signed on its own, with nothing linking one job to the next. EnCase and FTK are strong at copying and examining disks but print no certificate built for Indian evidence law. None of them cross from wiping into recovery.

“The parts exist. The combination does not.”

Q How big is the market?

About 650 million dollars of Indian demand is within reach, growing near 40 percent a year, on Deloitte and DSCI figures. We cite the published numbers rather than model our own.

Q Have you tested with a real forensic laboratory?

No. We could not get laboratory access in the window, and hackathon rules stop us leaving during the event. A pilot with one laboratory is the first item on our next-steps list, because it turns “it works on our laptops” into “it works on their evidence”.

The presenter's full set

Judges vary. Some know cryptography deeply, some are senior people from an earlier era of computing who care whether the thing is real and useful. Answer both without changing the facts.

For a judge who knows quantum computing

Q Your signatures are Ed25519. A quantum computer breaks that. Why should a court trust this in 2040?

It would, and that is exactly why we did not stop there. Every entry can also carry an **ML-DSA-65** signature, the scheme NIST standardised as FIPS 204 in 2024. It is built on lattice problems, which no known quantum algorithm solves efficiently. Ed25519 remains for today; the second signature is what survives.

Q Why does that matter more for forensics than elsewhere?

Because evidence has to stay checkable for decades. A case sealed today may be challenged in twenty years. Everywhere else, a broken signature scheme means you re-sign and move on. Here, the thing being protected is a claim about the past, and you cannot re-sign the past.

"An integrity guarantee with a known expiry date is not preserved integrity."

Q What about the hashes? Grover's algorithm.

Grover gives a square-root speed-up, so SHA-256 keeps roughly 128 bits of collision resistance against a quantum attacker. That is still comfortably sufficient, and NIST's own guidance treats SHA-256 as adequate. The urgent problem was the signatures, not the hash, and that is where we spent the effort.

Q So is it quantum-proof?

No, and we do not say that. We say the signature layer is quantum-ready: the scheme is standardised, it is attached additively so nothing breaks, and stripping it is detectable. Anyone claiming a whole system is quantum-proof today is overselling.

For computer science judges

Q Why length-prefix the fields? Why not just join them?

Because joining with a separator is not injective. Two different records can produce the same bytes to sign, which is a forgery primitive rather than a formatting choice. Every field carries its own four-byte length first, so the encoding is unambiguous. There is a regression test for exactly this and it must never be deleted.

Q Why implement verification three times?

Two implementations by one author can share a misreading of the specification and agree while both are wrong. A third, in a language with different string and integer behaviour, catches that class of error. A test asserts all three produce identical fingerprints and identical verdicts.

Q What is the complexity of verification?

Linear in the number of entries: one hash and one comparison each. Measured at 37 milliseconds for 500 entries, and it reports the first failing sequence number rather than a boolean.

Q You check the co-signer before wiping. What about the gap between the check and the write?

A genuine race, and we do not pretend otherwise. No tool that destroys data can close it: something can always change between the check and the irreversible act. We narrow it to a single operation, mark that operation degraded when it happens, and state it rather than deny it.

Q Why not a blockchain?

A distributed ledger solves disagreement between many mutually distrusting parties. Here there are two parties who both want a correct record, and the evidence must never leave the laboratory. A hash chain with an independent co-signer gives the integrity property without publishing case data or requiring a network.

For senior judges who want to know it is real

Q Explain it without the jargon.

Think of a bound register with numbered pages, where each page repeats a summary of the previous page. Tear one out and the next page no longer matches. Today's tools issue loose certificates instead: take one away and the rest look perfectly fine. We also make a second officer, sitting at a different desk, countersign every page.

Q Who would actually use this, and for what?

A forensic laboratory examining a seized drive. A police cyber cell recovering deleted messages. A bank disposing of old drives that must be provably wiped. A defence establishment doing both. In each case somebody must later prove what was done, to somebody who was not in the room.

Q What does it cost to run?

Two ordinary laptops and one power socket. Competing tools charge thousands of dollars per person per year. Ours is open, runs offline, and needs no server, no subscription and no special hardware.

Q Is this a student project or something real?

It runs, it is tested, and the two-computer signing was proven on real hardware on 3 September 2026. It is a working prototype, not a product: it has no government certification yet and has not been piloted in a laboratory. We say that plainly because the alternative is being caught overstating it.

Market, competitors, and what makes us different

WHO	WHAT THEY DO WELL	WHAT THEY CANNOT DO
Blanco, BitRaser	certified wiping, trusted, established	each certificate signed alone; no link between jobs; no recovery side
EnCase, FTK	imaging and analysis at scale	no certificate built for Indian evidence law; foreign, with export exposure; costly per seat
Autopsy (free)	capable open-source carving	no signed chain, no certificate, no second party
Akhanda	both halves in one signed register, two parties, statute-native certificate	not certified yet, not laboratory-piloted yet

WHY THE ESTABLISHED PLAYERS CANNOT SIMPLY COPY THIS

Blancco and BitRaser hold government certification. Changing the certificate structure means re-certification, which takes years and money, so a certified vendor moves slowly by design. EnCase and FTK are foreign products, and building a certificate native to one country's evidence law is poor economics for them. And none of them cross the line between wiping and recovery, because those are separate product lines with separate buyers inside their own companies.

The hostile questions

Q This is just a hash chain. Where is the innovation?

The hash chain is the easy part and we do not claim it. The innovation is the conjunction: erasure and recovery in one register, a second party on a physically separate machine, a certificate native to Indian evidence law, and a tool that blocks and records the block rather than downgrading itself. Each piece exists somewhere. The combination does not.

Q Your record cannot prove somebody did not do something off the books.

Correct, and we say so on the certificate itself. It proves nothing was *changed* after being written. It cannot prove nothing was *withheld*. Recording refused attempts narrows the gap; nothing closes it. No register can.

Q What if I unplug the second laptop?

The tool stops, and writes down that it stopped. That is the entire design. If a missing witness simply meant a weaker record, unplugging it would be a way to do unwatched work. Because the refusal is recorded, unplugging it becomes a way to leave a mark rather than avoid one.

Q Did AI write this?

We used AI tooling while building, as most teams now do. The design decisions, the honesty rules and the architecture are ours, and we can defend any line of it. The product itself calls no AI at runtime: it is deterministic, works offline, and you can check that with a text search.

NEVER SAY THESE

- “Legally guaranteed” — say the certificate contains what the section requires.
- “Unhackable” or “quantum-proof” — say quantum-ready signature layer.
- “Fully tested” — three things are marked not tested and we name them.
- “Blockchain” loosely — it is a hash chain with an independent co-signer, and the difference matters to anyone who knows.
- Any invented case number, officer name or device serial. Everything we show is real output.